

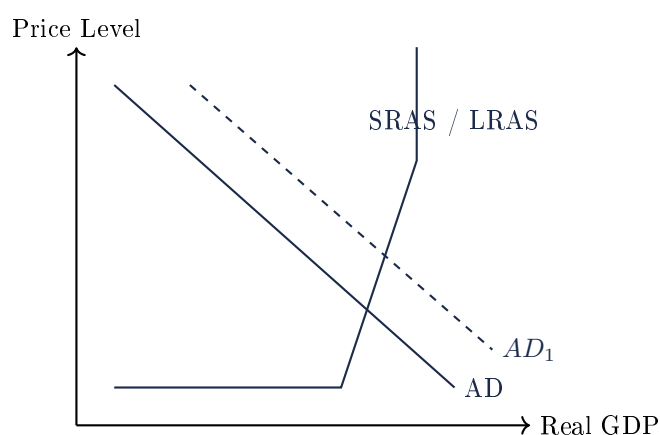
Document 11. AD-AS Framework & National Income

Topic overview. The AD-AS framework explains short-run and long-run movements in real national income and the general price level. It is the workhorse for the four macro issues and all macro policy questions.

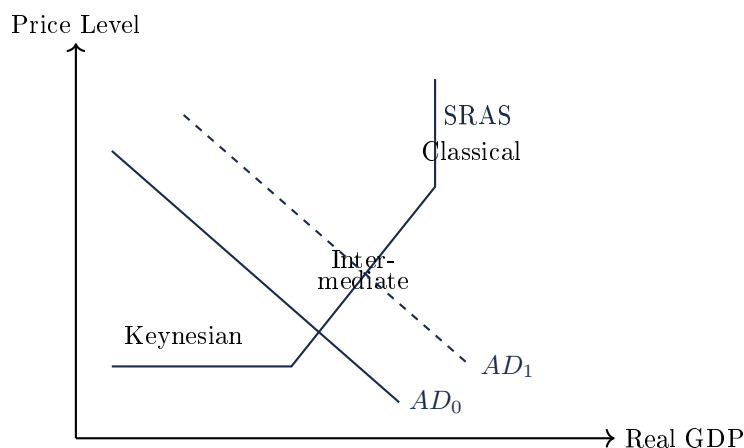
Section A. Important Definitions

Aggregate Demand (AD)	Total planned spending on domestic output: $AD = C + I + G + (X - M)$.
Determinants of C	Disposable income, wealth, interest rates, consumer confidence, expectations.
Determinants of I	Interest rates, business confidence, technological change, corporate tax rates.
Determinants of G	Discretionary fiscal stance; automatic stabilisers.
Determinants of $(X - M)$	Foreign income, exchange rate, relative inflation, trade policy.
SRAS	Short-Run Aggregate Supply. Upward-sloping; firms respond to higher prices in the short run.
LRAS	Long-Run Aggregate Supply. Vertical at full-employment output Y_f ; depends on productive capacity.
Multiplier k	$k = 1/(1 - MPC) = 1/MPW$. Captures the magnified effect of an injection on national income.

Section B. Diagram — AD-AS with Keynesian SRAS

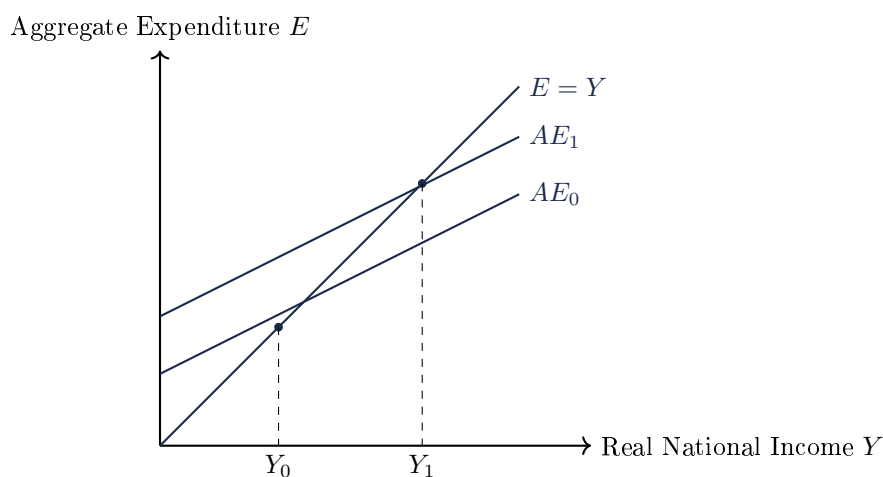


Section B1. Diagram — AD shift across the three SRAS ranges



Caption. In the Keynesian range, $AD \uparrow$ raises Y with no P rise. In the intermediate range, both Y and P rise. In the classical range (at Y_f), only P rises.

Section B2. Diagram — Multiplier (45-degree, Keynesian Cross)



Caption. A rise in autonomous expenditure shifts $AE_0 \rightarrow AE_1$, raising equilibrium income from Y_0 to Y_1 . $\Delta Y > \Delta AE$ due to the multiplier.

Section C. Question Type 1: Explain how a change in AD affects national income

Step 1. Identify the relevant component (C , I , G , or $X - M$) and explain the trigger.

Step 2. AD shifts (rightward / leftward) from AD_0 to AD_1 .

Step 3. Apply the multiplier: $\Delta Y = k \times \Delta \text{injection}$.

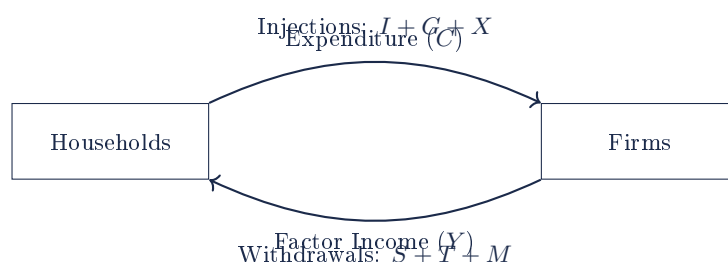
Step 4. Identify whether the economy is in the *Keynesian range* (flat SRAS, Y rises with no inflation), *intermediate range* (both Y and P rise), or *classical range* (at Y_f , only P rises).

Step 5. State the impact on macroeconomic objectives (growth, inflation, unemployment, BOT).

Section D. Question Type 2: Explain how a change in AS affects national income

- Step 1.** Distinguish SRAS (cost-driven) shifts from LRAS (capacity-driven) shifts.
Step 2. SRAS rightward shift: lower input costs, weaker currency, productivity gain. SRAS left: cost-push (oil, wages, tax).
Step 3. LRAS shift: changes in factor endowments, technology, productivity, labour-force size.
Step 4. Trace the new equilibrium: change in Y , P .
Step 5. Conclude on macro objectives.

Section E. Circular Flow of Income (4-sector)



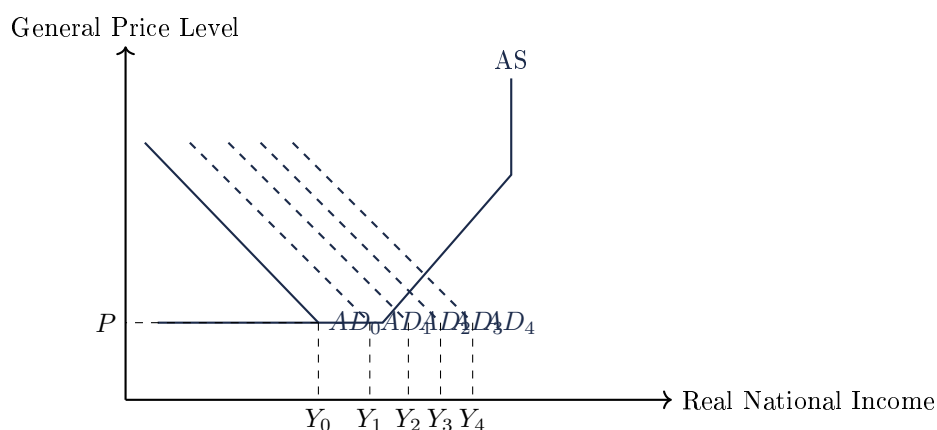
Equilibrium condition. $S + T + M = I + G + X$.

Section F. The Multiplier Process — Crucible 7-Step Framework (verbatim)

Multiplier numerical table (illustrative, $MPC = 0.6$, $\Delta J = \$100m$):

Round	Δ National Income	Induced Δ Consumption	Δ Withdrawals
1	\$100m	\$60m	\$40m
2	\$60m	\$36m	\$24m
3	\$36m	\$21.60m	\$14.40m
...	...	...	...
Total	\$250m	\$150m	\$100m

Diagram — Multiplier Process Using AD-AS Analysis (Crucible)



Caption. Successive rounds of induced consumption shift AD rightward by smaller and smaller amounts ($\$100m \rightarrow \$60m \rightarrow \$36m \rightarrow \dots$). On the Keynesian range, each round raises Y without raising P . Total $\Delta Y = k \cdot \Delta J$.

Step 1. Assume a four sector economy that is initially in equilibrium with total withdrawals (W) = total injections (J). The increase in $I/G/X$ represents an increase in J into the circular flow of income and the economy is no longer in equilibrium as J now exceeds W .

Step 2. For example, this could have resulted from increased government spending on infrastructure projects / increased technological investments by firms to increase profits / increased export revenue due to depreciation of the exchange rate.

Step 3. Assuming that the economy has excess capacity, an injection of \$100m of $I/G/X$ generates \$100m of income for households employed by firms in the construction / technological / export industry. As these firms face a fall in planned inventories, they pay households factor incomes in the form of wages, rent, interest and profits in exchange for factor services like labour, land, capital and entrepreneurship respectively to increase their output.

Step 4. These households will spend a proportion of the additional \$100m in income based on their marginal propensity to consume (MPC). Assuming that $MPC = 0.6$, an additional \$60m will be spent on domestically produced goods and services, while the remaining \$40m will be leaked out of the economy as additional savings (S), taxes (T) and imports (M).

Step 5. As one man's spending is another man's income, the induced consumption creates an additional \$60m of income for households employed in the consumer goods industry. These households will also spend a proportion of the additional \$60m (specifically, \$36m in this example) on domestically produced goods and services while the remaining \$24m is leaked out as additional S , T , and M .

Step 6. After each successive round, the additional induced consumption generated becomes smaller and smaller due to leakages such as S , T and M . This cycle of spending and re-spending continues until the total increase in leakages equals to the initial injection. Thus, the multiplier effect does not continue indefinitely.

Step 7. Eventually, the initial increased investment of \$100m creates a more than proportionate increase in real national income of \$250m. Hence, the multiplier is 2.5 in this example. Based on the formula $k = \frac{1}{1-MPC}$, we see that the smaller the MPC, the greater the size of the multiplier.

NOTE: Framework reproduced word-for-word from the Crucible Education Centre handout. For Singapore, k is small ($\sim 1.0-1.4$) because MPM and MPT are large in an open economy with high import content.

Section G. Singapore Application

- **Small k .** Singapore's openness limits the domestic multiplier; fiscal injections leak abroad through imports. This weakens the case for purely demand-side fiscal expansion.
- **Composition of AD.** C is small ($\sim 35\%$ of GDP), $X - M$ is large; hence external demand shocks dominate macro fluctuations.
- **COVID-19 fiscal response (2020).** S\$92 billion across five Budgets, roughly 20% of GDP, drew down past reserves with Presidential approval.

Section H. Worked Examples

Worked Example 1: Multiplier computation

Q. If $MPC = 0.6$, $MPT = 0.15$, $MPM = 0.25$, compute the multiplier.

$MPW = 1 - MPC = 0.4$. Hence $k = 1/0.4 = 2.5$. A \$100m injection raises Y by \$250m.
Note: MPT and MPM are already embedded in $1 - MPC$; do not subtract again.

Worked Example 2: Cost-push inflation shock

Q. How does an oil price shock affect Singapore's macro objectives?

SRAS shifts left. Equilibrium Y falls, P rises = **stagflation**. Cyclical unemployment rises; trade balance worsens via higher import bill. MAS tightening (S\$NEER appreciation) offsets the imported price effect but cannot restore Y ; supply-side and targeted transfers (U-Save, household subsidies) complement.